

# Introduction

## Making

Making is not new, but it is having a moment. That moment is the global maker movement that arose at the turn of the millennium. The beginnings of the contemporary movement are often associated with the first publication of *Make: Magazine* in 2005 by the entity now known as Make Community LLC and formerly known as Maker Media Inc., an investor-funded company that went out of business in June 2019 and was reconstituted as a member-funded organization in July 2019 (Dougherty 2019). Attendance at maker faires, Make Community–branded conventions featuring local makers, increased from 22,000 in 2006 to more than 1.4 million in more than two hundred cities worldwide in 2016 (Maker Media 2017). Many more cities across the world are home to **makerspaces**, which are workshops that offer access to and training on fabrication technologies like 3D printers and laser cutters, often with a focus on education and entrepreneurship. In the United States, President Barack Obama highlighted the maker movement when the White House hosted a maker faire in 2014 and a National Week of Making in 2015.

it is also notable to me that there is an increase in businesses that also cater to makers -- I did not expect such a focus on makers at a franchise like MicroCenter, for example

In other words, by 2014, the contemporary maker movement was well established, and multiple makerspaces had materialized (and dematerialized) in Seattle, where I lived at the time. But “makerspace” was a new word for

me when I heard it in a conversation at an academic conference in October 2014, and I used my phone to surreptitiously search the web for a definition because it seemed like a word I should already know. Later that same evening, I searched the web again, this time for “Seattle makerspace,”<sup>Page 2 →</sup> because I wanted to know if I could find one near me. The first promising result from my Google search was SoDo Makerspace, located in Seattle’s South of Downtown (SoDo) neighborhood. I was intrigued, but also intimidated. After all, I had only just learned a word that many other people seemed to already know, and I had no background in engineering or design. For most of my life, I had thought of myself as a writer—it was always my focus in academic work, and all my jobs after graduating from college had been about writing. How would a writer fit into a makerspace?

I finally visited SoDo Makerspace a few months later, in February 2015. I arrived on a Saturday for an open tour advertised on the website. When I arrived and met Eric Renn, the founder and CEO of the makerspace, I was caught off guard by his first question: “So, what do you want to make?” I couldn’t answer the question, because I had no idea what I could make. I left the tour dazzled by all the machines and objects I’d seen, and I was especially preoccupied with Eric’s question. Despite my struggle to answer it in the context of an unfamiliar space with intimidating machines, that question—*What do you want to make?*—resonates with questions that writers, writing teachers, and writing researchers have been asking for a long time.

I began this book project in an attempt to answer Eric's question, along with my own questions about making, the maker movement, and makerspaces. In 2015, I set out to conduct a longitudinal, ethnographically informed case study of a Seattle makerspace in order to gain a rich understanding of a local instance of this global phenomenon. I wanted to understand what (and whose) values and practices we are taking up and what boundaries we mark when we identify as makers or when we turn a writing classroom or a library space into a makerspace. The boundaries of what counts as "writing" and who counts as "writer"—particularly in relation to what counts as "making" and who counts as "maker"—have long been the concern of scholars across several areas of rhetoric and writing studies, including cultural and feminist rhetorics, material rhetorics, digital rhetorics, multimodal composition, and professional and technical communication. These areas come from different epistemologies and theoretical traditions, but they share a recognition that boundaries between discursive and material (and between human and nonhuman) are negotiated and marked, rather than given.

This book takes up these conversations and focuses on relations and boundary-marking practices in acts of making—especially what counts as making and who counts as a maker. These boundaries are not fixed or  
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→absolute, but are marked (and re-marked) through relations and interactions among people, things, technologies, communities, meanings, spaces, and disciplines. In other words, boundary marks and relations both shape and are shaped by what gets made (and what doesn't), by whom, and for whom.

Analyzing and understanding these marks, then, can help researchers, teachers, and practitioners make informed decisions about why and how to engage with the maker movement in our communities.

Consider, for example, that the boundaries that separate a human maker from a 3D printer are re-marked in a moment when the human uses her hand to help the machine feed plastic filament into its extruder as the machine prints. A writer who has previously not been regarded as a maker experiences a re-marking of boundaries as he learns and proves he can operate 3D printers and, in turn, is recognized as a maker. A researcher who had never seen a 3D printer before she started studying a makerspace notices that her early field notes relate what people are saying about 3D printers rather than what they are doing with 3D printers, but as she learns and becomes more familiar, the boundaries of her observations are re-marked to focus more on actions and processes, and not just words. And when a makerspace feels like a “total dudefest” (in the words of a man who lamented that fact to the only woman present in a makerspace that day), or is inaccessible to a person with a disability, then the boundaries of what is possible are opened up for some bodies, but closed down for other bodies. In other words, despite the name “makerspace,” neither “maker” nor “space” is a given, but is continually and differently negotiated by different bodies. We must answer for these boundary marks in the theories, traditions, and practices of making we privilege in our communities, institutions, and classrooms.

the question of who gets to count as a “maker.”

Indeed, people who do not work in or interact directly with the maker movement are increasingly likely to encounter definitions of “making” and “maker” that are influenced by the maker movement in professional, community, and educational contexts. Public libraries and libraries in educational institutions are creating their own makerspaces, and the maker movement has attracted scholarly and pedagogical interest across disciplines and areas, including digital humanities, education, human-computer interaction, library and information science, business, and rhetoric and composition. In rhetoric and composition, my primary field, there is growing interest in making and makerspaces, as evidenced in increased representation in recent calls for proposals for conferences and collections and in the work of a growing body of researchers and teachers, including KrystinPage 4 → Gollihue (2019), Maggie Melo (2018), Jentery Sayers (2017), David Sheridan (2010), John Sherrill (2014), Jason Tham (2019), and Stephanie West-Puckett (2017).

While this project engages primarily with perspectives on and studies of the relatively new maker movement, there are long histories and traditions of making that precede contemporary trends. As John Hagel, John Seely Brown, and Duleesha Kulasooriya (2013) explain,

For millennia, people have been manipulating objects to suit their needs and transform the world around them. Our built world and the many inventions and innovations that populate it are testament to the long history of making. What is new is how modern technologies, globalization, and cultural shifts are enabling and motivating individuals to participate in making activities and removing barriers all along the value chain, from design and prototype to manufacturing to selling and distribution. (3)

Hagel, Brown, and Kulasooriya position the maker movement as the “next generation of inventing and do-it-yourself” (1), suggesting a lineage in twentieth-century hobby shop culture and the DIY movement of the late twentieth and early twenty-first centuries. Faythe Levine and Courtney Heimerl (2008) trace the maker movement through traditions of crafting and design beginning with the Arts and Crafts movement of the twentieth century. Jessamyn Hatcher and Thuy Linh Nguyen Tu (2017) more pointedly tie the contemporary maker movement to “the rise of neoliberalism, when problems of deindustrialization . . . were no longer addressed primarily through collectives—whether government, policy institutes, or unions—but instead devolved to individuals and businesses” (274). While the contemporary maker movement is situated in a specific technological, economic, and sociopolitical context, *making* resonates across time and contexts, as scholars of material culture like Tim Ingold have observed. Ingold (2013) defines making as “a process of correspondence: not the imposition of preconceived form on raw material substance, but the drawing out or bringing forth of potentials immanent in a world of becoming” (31). My study of a makerspace traces processes of becoming-through-making that include objects, bodies, technologies, relationships, and spaces.

making as a kind of  
becoming

In engaging with a local instantiation of the contemporary maker movement, this project builds on an emerging, cross-disciplinary body of work that examines the marking of difference in rhetorical practices of making. Page 5 →

At the heart of this work is an understanding that neither the maker movement nor the technologies and objects associated with it are neutral. Rhetoric and composition scholars like Cynthia Selfe and Richard Selfe (1994), Adam Banks (2006), Angela Haas (2012), Laura Gonzales (2018a), and Jody Shipka (2011) have argued that technologies and our bodies' interactions with them are culturally situated. As Shipka (2011) argues, "We need to consider what is at stake—who and what it is that we empower or discount—when we use the term ["technology"] to mean primarily, or worse yet, *only* the newest computer technologies" (21). Furthermore, digital humanities scholar Jentery Sayers (2017), emphasizes the importance of redressing "the normative assumptions and effects of popular maker cultures—usually white, cisgender, straight, male, and able-bodied"—through critical interventions in and practices of making (7). And Debbie Chachra (2017), a materials scientist and engineering educator, makes a case against identifying as a "maker" because the "cultural primacy of making, particularly in tech culture—that it is intrinsically superior to not-making, repair, analysis, and especially caregiving—is informed by the gendered history of who is credited with making things, and, in particular, who made things that were shared with the world, not merely for hearth and home" (319).

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In fact, by its own account—and despite efforts to the contrary from some within the movement—the maker movement skews heavily toward male-identified people. According to Maker Media's (2015) press fact sheet, 81%

of the readers of *Make: Magazine* identified as male. And 70% of attendees at the 2014 Bay Area Maker Faire identified as male. Similarly, in Hackster.io's (2016) maker survey (in partnership with technology giants such as Intel, Microsoft, Google, Amazon, and Arm) of more than three thousand people across 104 countries, fewer than 7% of respondents identified as women. (Interestingly, neither Maker Media nor Hackster.io reported on race/ethnicity.) Additionally, scholars like JooYoung Seo and Gabriela Richard (2018) and Katherine Steele, Maya Cakmak, and Brianna Blaser (2018) point out the ways in which makerspaces and maker technologies can exclude people with disabilities, particularly when people with disabilities are not included in the process of planning and designing these spaces. While some identify makerspaces and maker practices as opportunities to bring underrepresented groups, such as women, into science, technology, engineering, and mathematics (STEM) fields (Blackley et al. 2017), others point to women's exodus from male-dominated makerspaces and hackerspaces to form women-centric spaces that promote feminist ways of making and collaborating (Fox, Ulgado, and Rosner 2015). The attention of these scholars to the underrepresentation of women in the mainstream maker movement resonates with the observations of scholars like Anne Balsamo (2011) and Sasha Costanza-Chock (2020) who trace broader accounts of erasures and exclusions of women and marginalized communities in technological innovation and design. For these reasons, I am wary, like education scholars Shirin Vossoughi, Paula Hooper, and Meg Escudé (2016), of the "uncritical adoption of branded versions of making, particularly with regard to their implications for education equity" (210).



Building on this scholarship, my observations and experiences from a makerspace, and the perspectives of the people with whom I worked, this book models a relational framework for understanding and participating in acts of making. This framework operates at multiple levels: the making and remaking of knowledge across theories, perspectives, and experiences; the making and remaking of bodies, relationships, meanings, technologies, and the makerspace itself; and the making and remaking of practices across local maker communities, the global maker movement, and maker-inspired pedagogies. At each level, I highlight relations and boundary marks that are dynamic, collaborative, multimodal, material, and culturally situated. In other words, while this book is a study of acts of making in a makerspace, it is also a study of how the boundaries of “maker,” “making,” and “knowledge” both shape and are shaped by relations among bodies, technologies, traditions, materials, things, and spaces. As a project that works across disciplines and across theory and practice, the book offers implications and applications of this relational framework for people who identify as makers, as well as community strategists and technical communicators, and also for writing teachers committed to supporting students’ interests, backgrounds, and rhetorical skills.

# Chapter Overview

The first portion of the book articulates a relational framework for making that includes making knowledge *about* making. Chapter 1 situates rhetoric and/as making, and it assembles understandings of relationality that guide my work. This chapter also introduces **relational rhetorics** as a way of participating in and accounting for relationships with and among bodies, spaces, theories, perspectives, and experiences. The theories, methodologies, practices, and relationships in this project are as much a part of the study phenomenon as the acts of making I examine. Drawing on traditions Page 7  
→across epistemologies, Chapter 2 offers stories as a method/ology for relational rhetorics that offers not only my accounts and interpretations of experiences with makers and making, but also how I came to those accounts and interpretations and experiences. Readers who are interested in the dynamics and processes of qualitative research projects (like a longitudinal, ethnographically informed case study) will find in this chapter a detailed account of my philosophies, relationships, and techniques in conducting and writing about research on making.

The second portion of the book offers two chapters with stories from SoDo Makerspace. Readers interested in learning more about what I observed and experienced in my years studying a makerspace will find in these chapters stories about acts of making, rhetorical approaches to design and documentation, and dynamic relations among humans and machines in a makerspace over several years. These stories are my participation in (rather than my observation or representation of) the marking of boundaries. Enacting the method/ology described in Chapter 2, each story begins with a particular moment that came to matter to me and to the people I was

interacting with. The stories trace how those particular moments came to matter, not only through the prior interactions that informed my understanding of and participation in that moment, but also through the ways in which that moment changed how I understood past and future interactions. I also connect these stories with the theoretical and methodological concepts discussed in Chapters 1 and 2.

Chapter 3 is a story of **spatial manipulations** (a phrase inspired by a 3D-printed “spatial manipulation toy”): interacting with people and objects, making objects, making space in a makerspace, and understanding embodied ways of meaning and making. The chapter focuses on the ways in which my own efforts to make space for myself in (or integrate myself into) a makerspace are entangled with how I understand what I observed there, particularly the efforts of Tony, a freelance writer, to make space for himself in the makerspace. And while Chapter 3 focuses on making things, Chapter 4 traces the complex function of disequilibrium (which includes breaking things) in making processes in SoDo Makerspace: from human errors and machine malfunctions to the amateur ethos of the space (and the maker movement) and systems of production. Some makers allow for and even cultivate a relatively high tolerance for disequilibrium as an invention practice. This story traces disequilibrium as both a marking of boundaries *and* a site for intervening in boundary marking. The threads of this story examine and account for disequilibrium-as-boundary-marking in Page 8  
→machine operation and maintenance, in documentation practices, in prototyping and design, in the ethos of the space, and in production systems.

The third and final portion of this book looks beyond the physical space of SoDo Makerspace to trace relationships and practices of making. The relationships and ventures of makers are dynamic, and they exceed the physical and temporal space of a single makerspace, as Chapter 5 and the conclusion describe. In turn, my relationships and experiences in and beyond the makerspace shaped my approach to teaching writing and rhetoric, described in Chapter 6. Readers interested in the ways in which practices of making apply to professional and pedagogical work beyond makerspaces will find in these chapters discussions that illustrate these relationships and possibilities.

Chapter 5 focuses on professional practice in the maker movement and intersecting technology industries, and Chapter 6 focuses on iterations of a digital rhetorics course that was influenced by my research on making and relationships with makers. Both these chapters involve collaborative knowledge-making with my research participants and account not only for *what* we found in our collaborative analysis but also *how* we found what we found. In other words, just as Chapters 3 and 4 account for the making of my research(er) apparatus in SoDo Makerspace, these chapters account for the boundary-marking practices involved in collaborative knowledge-making in and beyond SoDo Makerspace, with implications for professional practice and for teaching. Thus, the methodology in this project can be useful not only for theorizing but also for (professionally) *doing* and *teaching* writing as complex, relational, material-discursive acts of making. Chapter 6 turns to implications for teaching by tracing the ways the concepts introduced in previous chapters shaped my approach to teaching writing and rhetoric. The stories in this chapter connect concepts like rhetoric as/and making,

disequilibrium, spatial manipulations, stories, design, and community across my research and teaching practices to show the ways in which they informed each other. Finally, in the conclusion, I reflect on the affordances and constraints of this study and articulate future directions for the study, practice, and teaching of making.

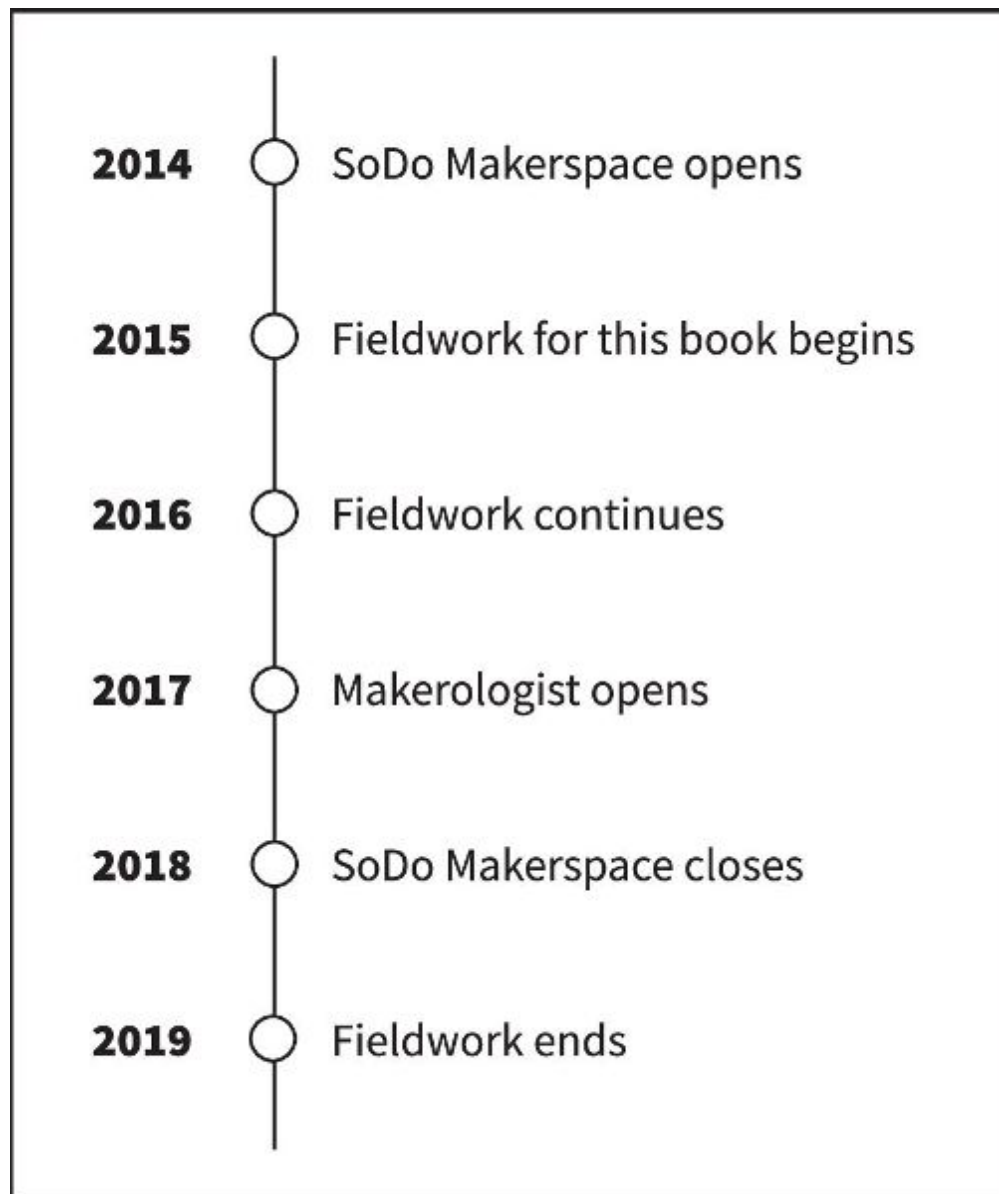


Figure 1. Chronology visualization: 2014–2019

# **Chronology Visualization: 2014-2019**

While the stories in this book model a nonlinear, relational approach to knowledge making and storytelling, the visualization in Figure 1 presents a chronological narrative that emerges from and can be read alongside the nonlinear relations in the chapters, which move forward and backward in time to emphasize relationality. This chronological visualization provides a temporal orientation to the stories in this book.